LITERATURE SURVEY

Phishing Website Detection

Using Web Crawled Content Dataset

Based on 15 Research Papers

Machine Learning & Deep Learning • LLMs • Focused Crawlers

Ensemble Methods • Blockchain Security • Adversarial Attacks

Overview of Literature Survey

15 research papers spanning 8 key thematic areas in phishing detection

01 LLM-Based Detection
GPT-4, LLaMA-3, Claude-2 for threat & defense

02 ML & Vision Methods
MobileNetV2, CNN-LSTM, Random Forest, SHAP

03 Focused Web Crawling
Crawl-shing, MCS algorithm, harvest rate +21%

04 Feature Frameworks
CASE, PILWD-134K dataset, semantic fusion models

05 Ensemble & Transfer Learning
CDEPF, RBE, cross-dataset accuracy 94.4%

06 Adversarial & Emerging Threats
Phishing 2.0, quishing, domain ownership abuse

07 Blockchain Security
B-PhishQR, QR code phishing, cryptographic audit

08 Research Gaps & Future Work
Open challenges and directions ahead

LLM-Based Phishing Detection (Papers 1 & 8)

Paper 1 — Systematic LLM Review

Models Studied
GPT-4, LLaMA-3, Claude-2

Threats Identified
Multimodal phishing, AI-generated spear-phishing

Key Finding
LLMs outperform traditional ML on nuanced linguistic cues but introduce new attack surfaces

Method
Quantitative publication trend analysis + head-to-head benchmarking vs. human experts

Paper 8 — DEMO Framework

Core Concept
Adversarial Asymmetry — attackers easily change perceptual surface (text/images) but leave inconsistencies in the infrastructural skeleton (URL/DOM)

Framework
DEMO: Decouples semantic identity claims from network evidence using LLMs

Data Sources
CTI from APWG & Microsoft reports anchored to real network inconsistencies

📊 Machine Learning & Vision Approaches (Papers 2, 9, 10, 11)

Paper 2 — AI-Driven CV & ML

- Architectures: MobileNetV2, CNN-LSTM, Random Forest
- SHAP identifies: Packet Header Length & Domain Entropy
- MobileNetV2 best for vision-based website image detection
- Focus: IoT / edge resource-constrained environments

Paper 9 — RBE Feature Selection

- Recursive Backward Elimination (RFE + BE hybrid)
- Reduces dimensionality without sacrificing accuracy
- Peak accuracy: 90.7% on email datasets
- Identifies key linguistic patterns & metadata

Paper 10 — Ensemble Transfer Learning

- PCA for feature harmonization across datasets
- CDEPF algorithm — Cross-Domain Ensemble Probability Fusion
- Cross-dataset accuracy: 94.4% (UCI → PhiUSIIL)
- Addresses temporal distribution shift problem

Paper 11 — Self-Supervised Pre-training

- SSL learns from large unlabelled corpora
- 2D syntactic features (CV) + contextual NLP features
- Masked prediction for semantic representation
- Outperforms purely contextual models

Focused Web Crawling & Dataset Construction (Papers 4, 7, 12)

Paper 4 — Crawl-shing Focused Crawler

- Graph Isomorphism + Maximum Common Subgraph (MCS) for semantic similarity
- **21% better harvest rate vs standard BFS crawlers**
- Retrieves phishing-feed pages beyond keyword-based VSM crawlers

Paper 7 — PILWD-134K Dataset

- 134,000 verified phishing & legitimate samples
- Includes URLs, HTML, screenshots & technology fingerprints
- **27 novel features: phishing sites are 'effortless clones' with fewer active tech stacks**

Paper 12 — PhishXtract: Longitudinal Domain Ownership Study

Attacker-Owned

Newly registered malicious domains — traditional proactive detection works here

Compromised Domains

Legitimate sites hijacked by attackers — harder to blocklist proactively

Third-Party Hosted

Abuse of platforms (cloud, CDN, SaaS) — evades new-registration detection entirely

🔍 Feature Engineering Frameworks (Papers 5 & 6)

Paper 5 — CASE Framework & Multistage Model

C — **Counterfeiting**
Brand impersonation signals

A — **Affiliation**
Links to malicious infrastructure

S — **Stealing**
Credential-harvesting patterns

E — **Evaluation**
Risk assessment & scoring

3-Stage Detection Pipeline

① Whitelist Filter → ② Fast Favicon Check → ③ ML Classification

Paper 6 — Multi-scale Semantic Fusion Models

- URL
- Title
- Body Text
- Invisible Text

Fusion Models Proposed:

MDF	Multi-scale Data-layer Fusion	F1: 0.971
MFF	Multi-scale Feature-layer Fusion	F1: 0.978
MIF	Multi-scale In-depth Fusion	F1: 0.983 ★ BEST

Internal semantic structure > visual elements for detection

⚠️ Emerging & Specialized Threats (Papers 3, 13, 14, 15)

Paper 3 — Ethereum Fraud Detection

- Phishing & Ponzi schemes on blockchain
- Methods: XGBoost, deep learning, graph-topology features
- Features: transaction data, smart contract opcodes
- Network graph analysis reveals fund-flow anomalies

Paper 13 — Phishing 2.0 & Human Limits

- GenAI creates hyper-personalized phishing content
- Human detection accuracy: only 57% for AI content
- Media tested: portraits, emails, SMS, logos
- AI social engineering is increasingly indistinguishable

Paper 14 — Arabic Phishing Detection

- Arabic_Fraud_Phishing (AFP) dataset — translated corpora
- Optimized ensemble stacking: LR, Naïve Bayes, AdaBoost
- Full vs binary stacking comparison
- Handles linguistic variances unique to Arabic phishing

Paper 15 — B-PhishQR (QR Phishing)

- Combats 'quishing' — QR code phishing attacks
- Blockchain-inspired cryptographic immutability
- Architecture: Blockchain Server + Key Mgmt + Verifier
- MFA integration + manual revocation of bad QR codes

📈 Comparative Analysis of Approaches

Paper	Approach	Key Method	Best Metric	Dataset
P1	Systematic LLM Review	GPT-4 / LLaMA-3	—	Multiple
P2	Vision + ML (IoT)	MobileNetV2, SHAP	~97% acc.	Custom IoT
P5	CASE Feature Framework	3-stage multistage pipeline	Realistic ratio 20:1	CASE dataset
P6	Semantic Fusion (MIF)	URL+Title+Body+HTML fusion	F1 = 0.983	Web crawled
P7	PILWD-134K Dataset	Web tech fingerprinting	27 features	134K samples
P9	RBE Feature Selection	RFE + Backward Elimination	90.7% acc.	Email corpus
P10	Ensemble Transfer Learning	CDEPF + PCA	94.4% acc.	UCI+PhiUSIIL
P11	Self-Supervised Learning	SSL + CV + NLP masking	SOTA	Unlabelled

★ *Paper 6 (MIF) and Paper 10 (CDEPF) show strongest generalisation across datasets*

Research Gaps & Future Directions

Dataset Imbalance & Realism

Most datasets have unrealistic phishing ratios. Real-world ratios (~20:1 legitimate:phishing, per Paper 5) are rarely replicated, skewing model performance estimates.

LLM-Augmented Phishing (Phishing 2.0)

Human detectors succeed only 57% of the time (Paper 13). Systems must counter AI-generated hyper-personalised attacks that bypass traditional rule sets.

Abuse of Legitimate Infrastructure

Paper 12 reveals attackers increasingly abuse third-party hosting. New-domain-only proactive detectors miss the majority of modern phishing deployments.

Cross-Domain & Multilingual Detection

Paper 14 shows Arabic phishing has unique linguistic patterns. Generalised multilingual models and improved cross-dataset transfer (Paper 10) remain open challenges.

Novel Vectors: QR & Blockchain Attacks

Quishing (Paper 15) and Ethereum fraud (Paper 3) demonstrate that detection must expand beyond web URLs to encompass QR codes, smart contracts, and on-chain activity.

Real-Time Edge Detection

Paper 2 highlights the need for lightweight models deployable on IoT/edge hardware. A gap remains in zero-latency, resource-constrained phishing detection pipelines.

CONCLUSION

Phishing Website Detection
Literature Survey

- ML & DL approaches (MIF, MobileNetV2, CDEPF) deliver strong detection accuracy with F1 scores exceeding 0.98 on benchmarks.
- Semantic content from web crawled data (URL, HTML, body text) outperforms purely visual features for phishing classification.
- LLMs (GPT-4, LLaMA-3) are a double-edged sword — enabling sophisticated attacks and advanced defences simultaneously.
- Novel threats (quishing, blockchain fraud, AI-generated phishing) require beyond-URL multi-modal detection strategies.

Future work must focus on cross-domain generalisation, multilingual datasets, real-time edge inference, and combating Phishing 2.0.